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Getting started with Secure manager

Tomas Team

- Timing : 30 minutes
- Purpose : Install the secure manager with focus on functionalities associated
- Process :
 - Introduction to TrustZone protection and new PRODUCT_STATE
 - Install the secure manager and an user application demo which use it.
- Payoff :
 - Ability to install Secure manager and see how simple it is
 - Understand what you can do with secure manager

Acronyms

Acronyms used in this workshop

- API : Application Programming Interface
- TZ: TrustZone
- TEE : Trusted Execution Environment
- PSA : Platform Security Architecture (dedicated to IOT hardware)
- SMAK : Secure Manager Access Kit
- SMDK : Secure Module Development Kit

Tools installation review

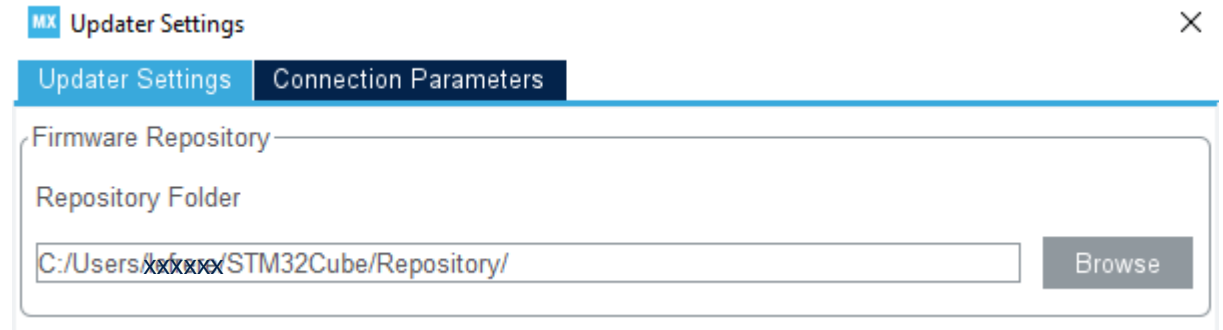
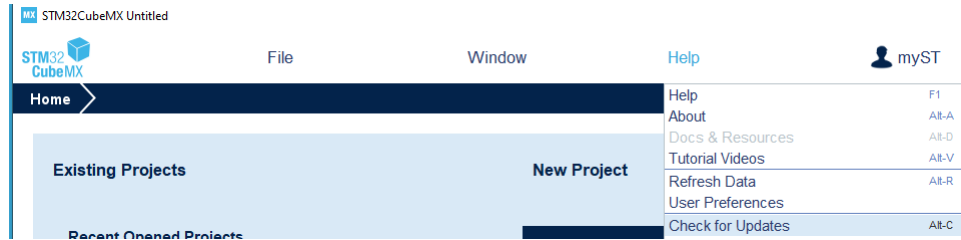


Tools installation

- SW Requirement :
 - [STM32CubeMX](#) version 6.9.2 or upper 
 - [STM32CubeH5](#) version 1.1.1 
 - [STM32TRUSTEE-SM](#) version 1.1.1 
 - [STM32CubeIDE](#) version 1.13.2 or upper 
 - [STM32CubeProgrammer](#) version 2.14.0 or upper 
 - [Material for hands-on](#) 
 - Terminal application ([TERATERM](#)) 



Issue with space in repository path...



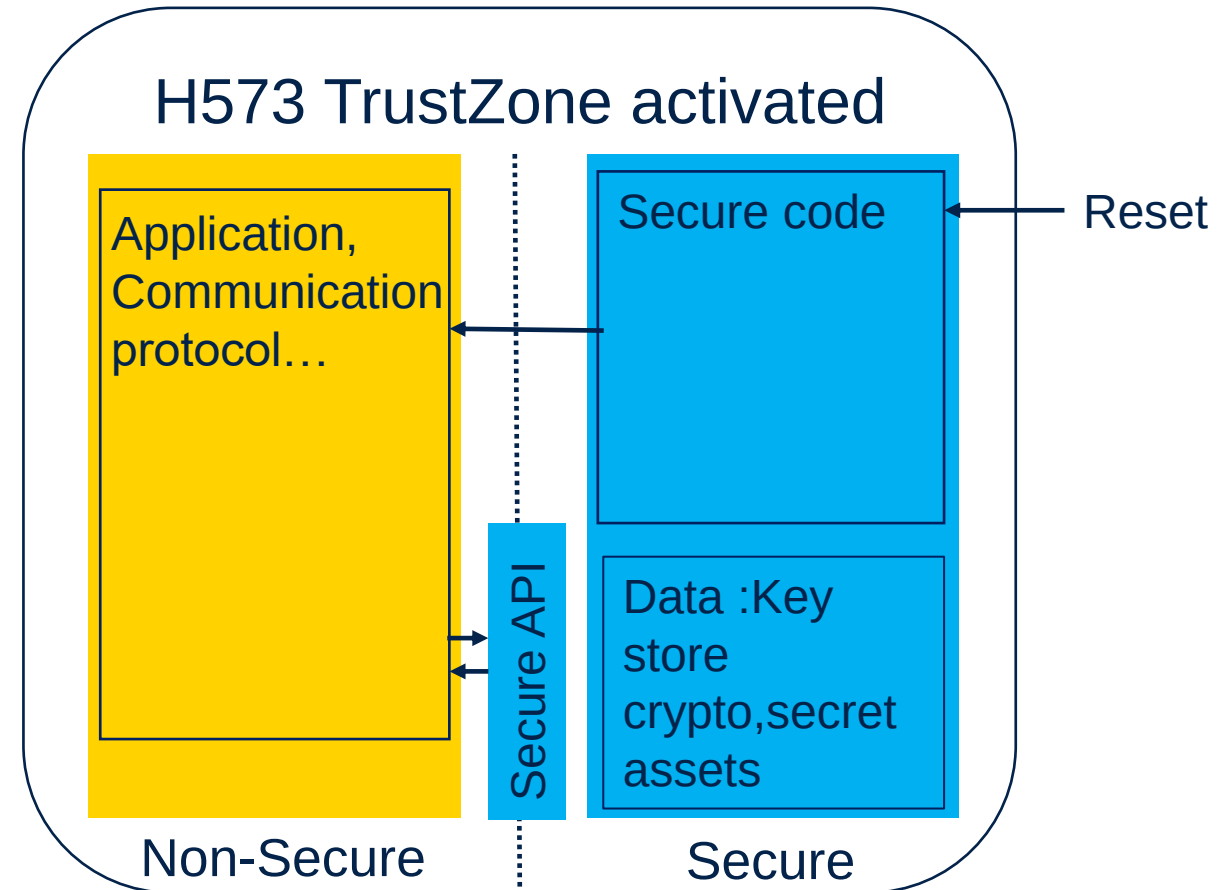
- By default the repository is :
C:\Users\<user name>\lefrere\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1
- If you have space in you <user name>, you will have issue....
- Workaround copy STM32Cube_FW_H5_V1.1.1 to a location without space
For example :
C:\Training\STM32H5_Workshop

Introduction to TrustZone protection and new PRODUCT_STATE



TrustZone isolation in 5 minutes...

- Introduced by ARM in Cortex-M23 and Cortex-M33
- Adds new internal state : Secure / Non secure (Trusted / Non Trusted)
- Hardware isolation between trusted firmware and non trusted
- In trusted part, usually: Key store, crypto, secret assets
- In untrusted part: Application, communication protocols
- Always boot in secure : trusted application configures the platform: decides split of memories and hardware IPs between secure and non secure
- Jumps to non secure application
- Secure provides an API to access to secure services



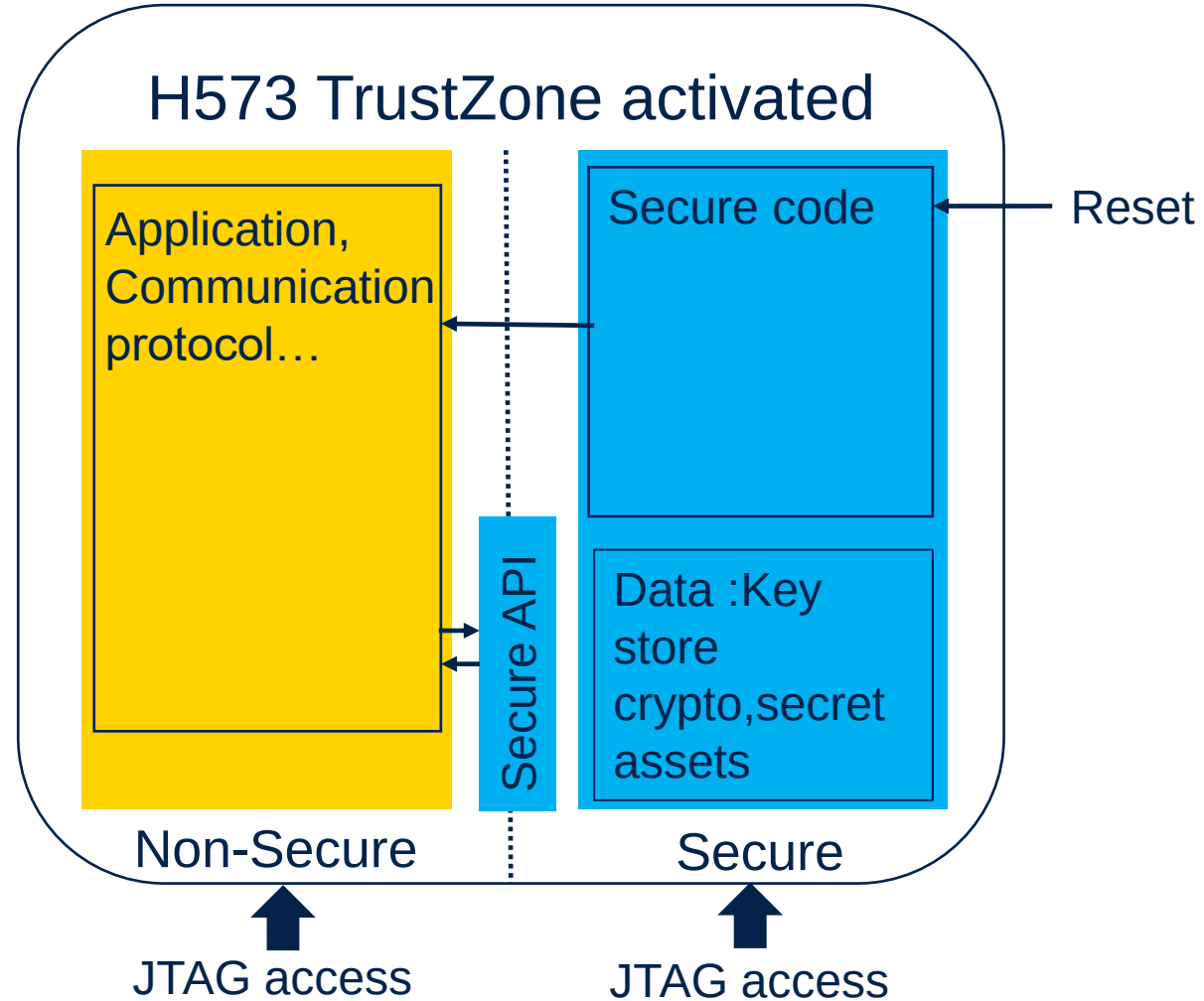


Product state introduction

- STM32H5 introduces PRODUCT_STATE (as of RDP for other STM32 families)
- **OPEN** state (equivalent to RDP0) : used for development
- **CLOSED** state (equivalent to keyed RDP2) : No JTAG/SWD access. Allows regression
- **LOCKED** state (equivalent to RDP2) : No JTAG/SWD access. No regression
- **TZ-CLOSED** (equivalent to RDP 0.5) : Only secure memory is protected. Allows development in non secure

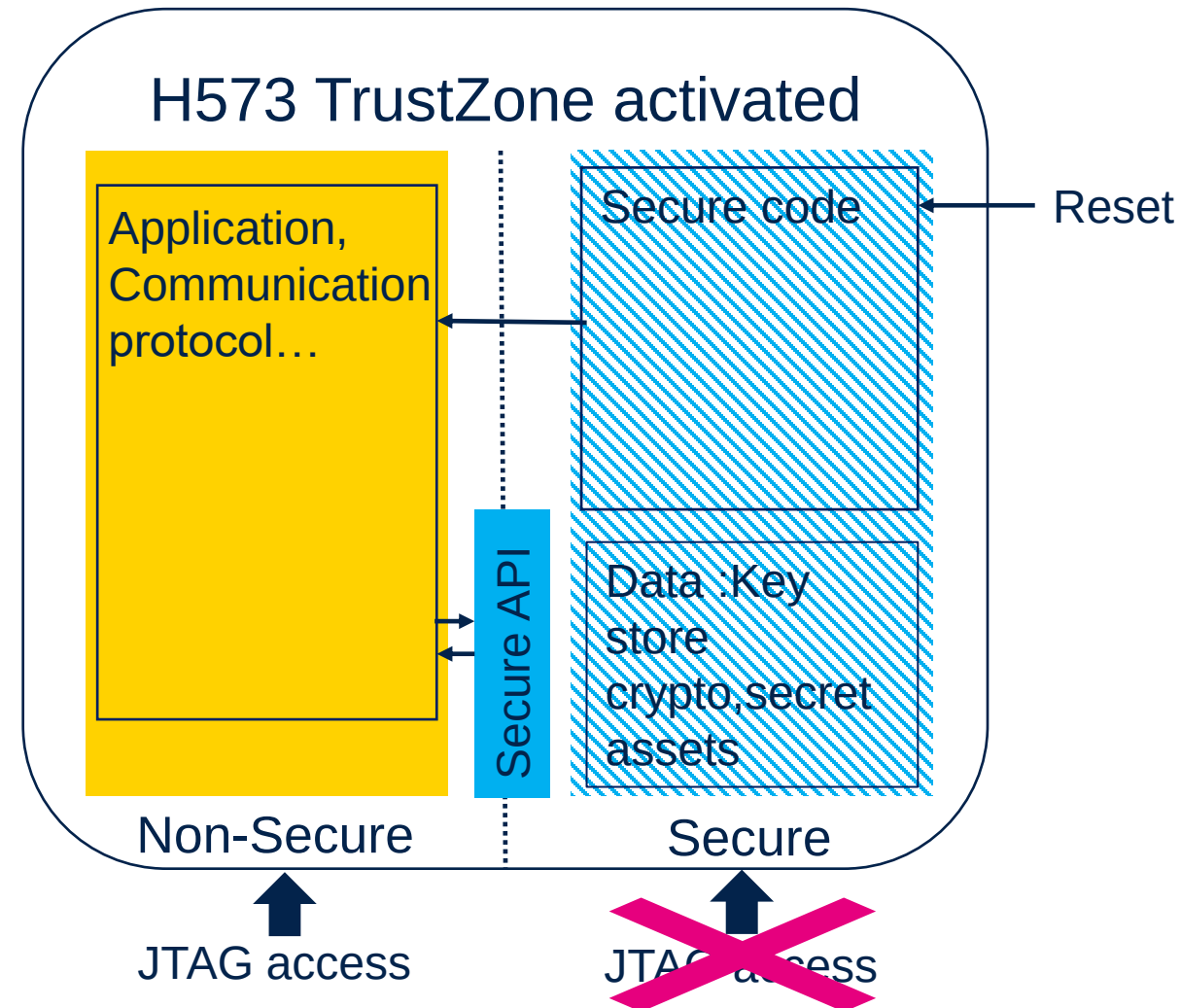


Product state OPEN (equivalent RDP 0)



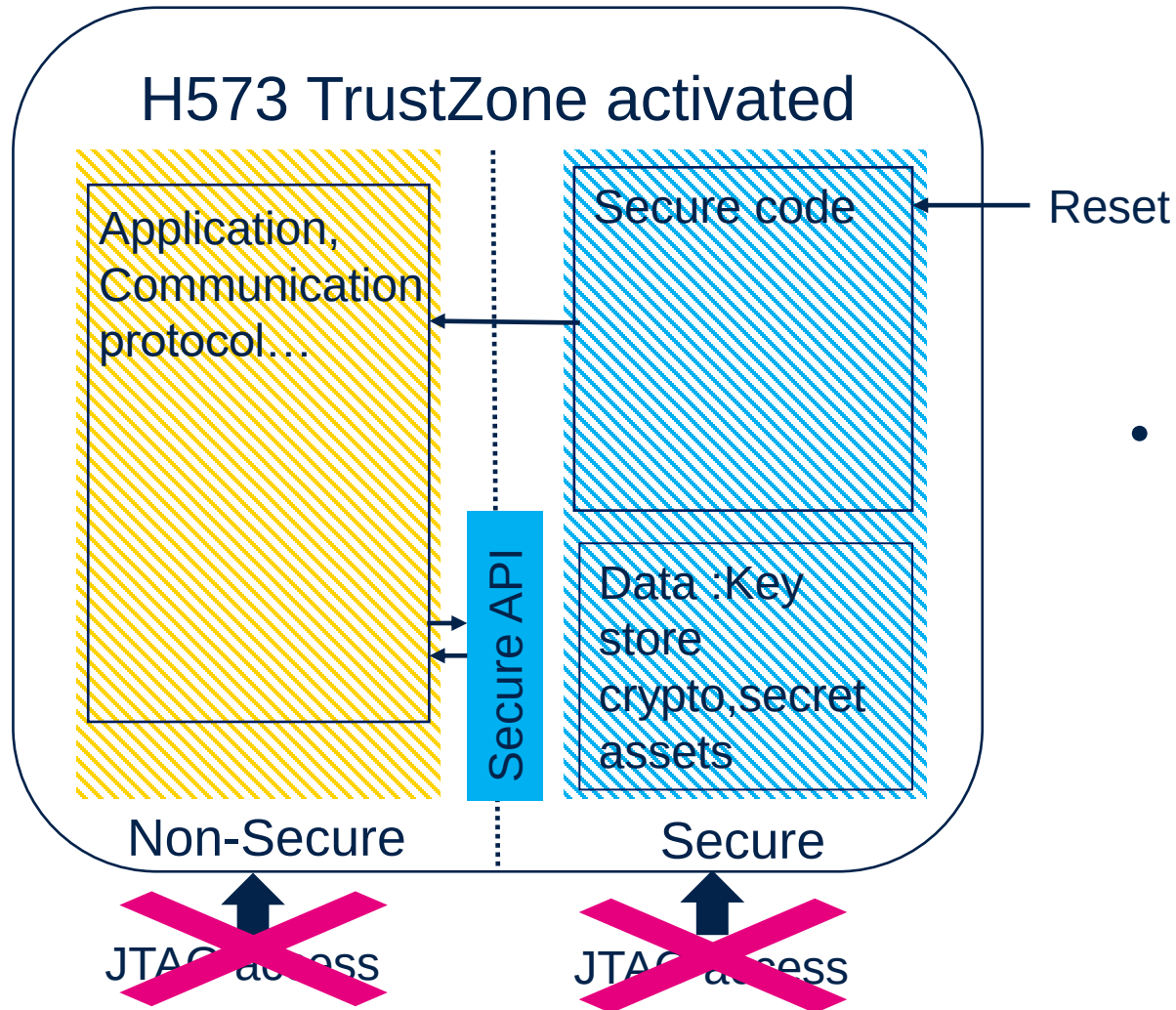


Product state TZ-CLOSED (equivalent RDP 0.5)





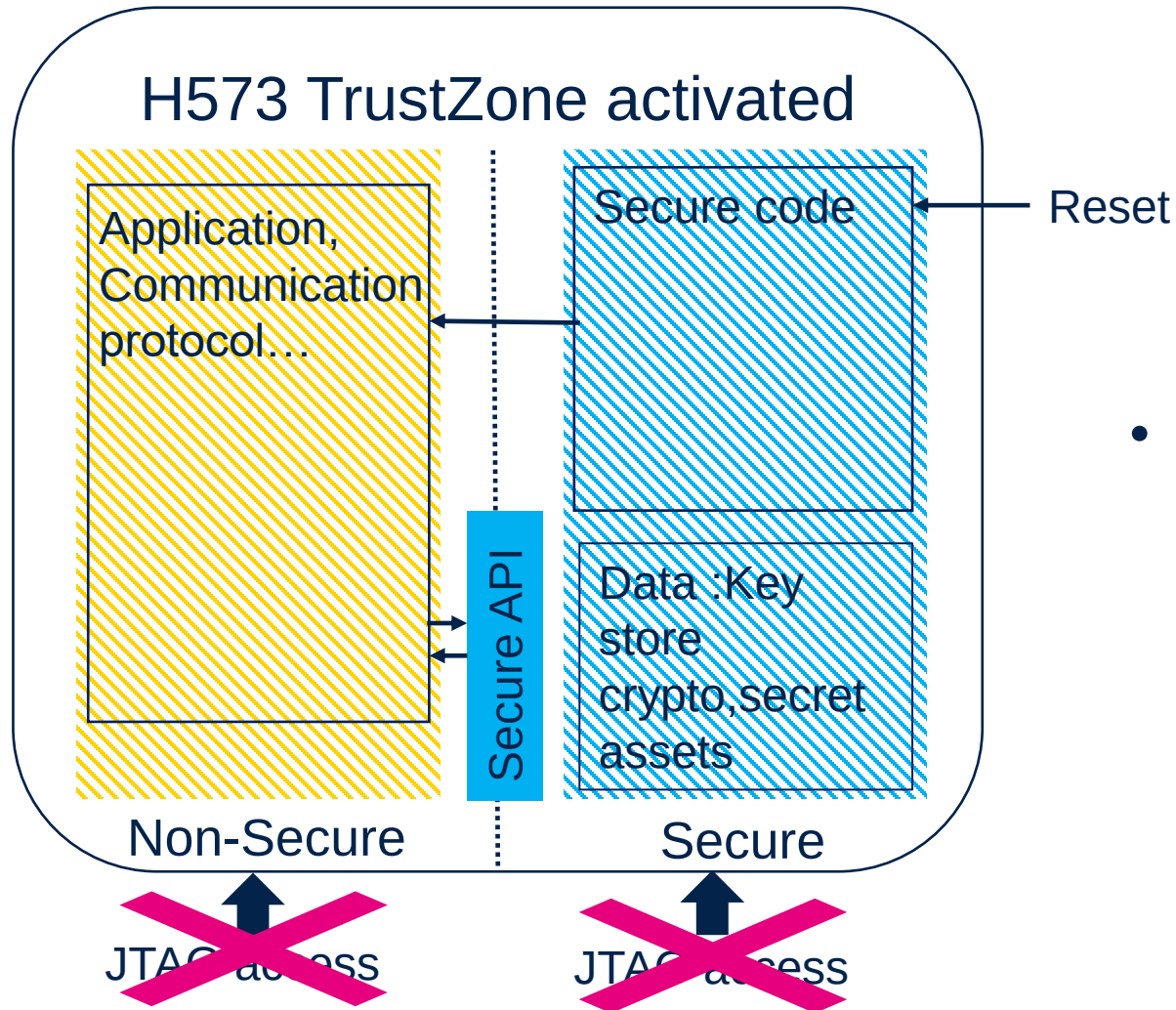
Product state CLOSED (equivalent RDP 1)



- Possibility come back to OPEN state associated with a fullchip erase



Product state LOCKED (equivalent RDP 2)

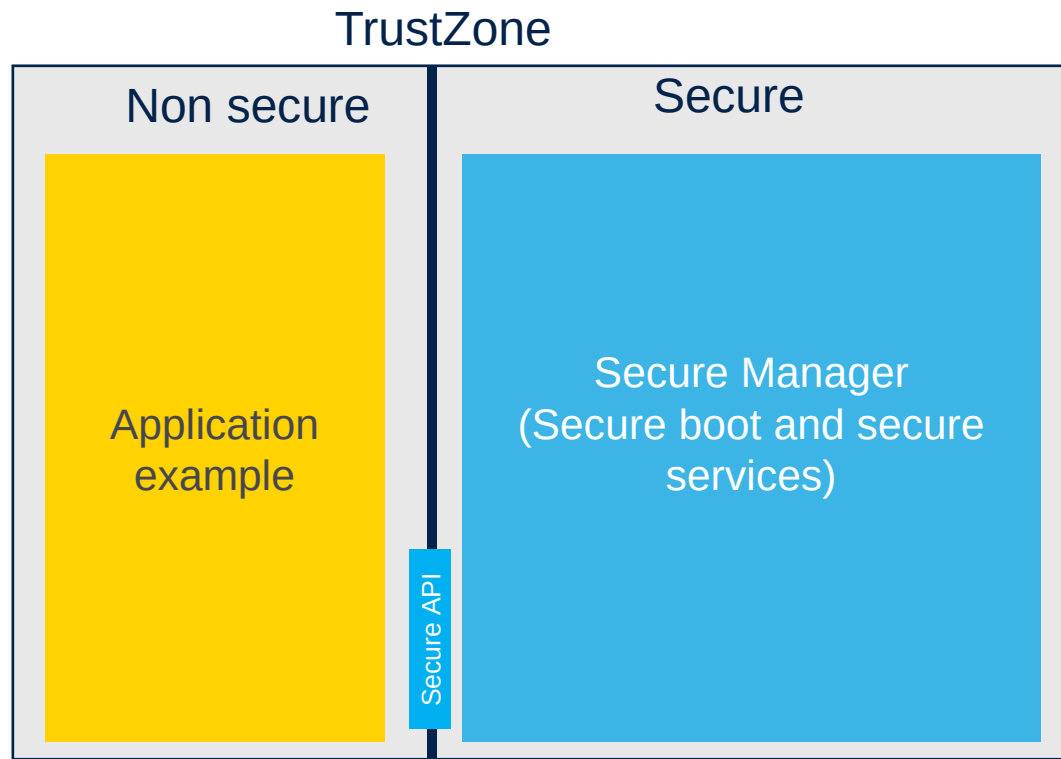


- No possibility to reopen the device

Ready to discover Secure Manager !



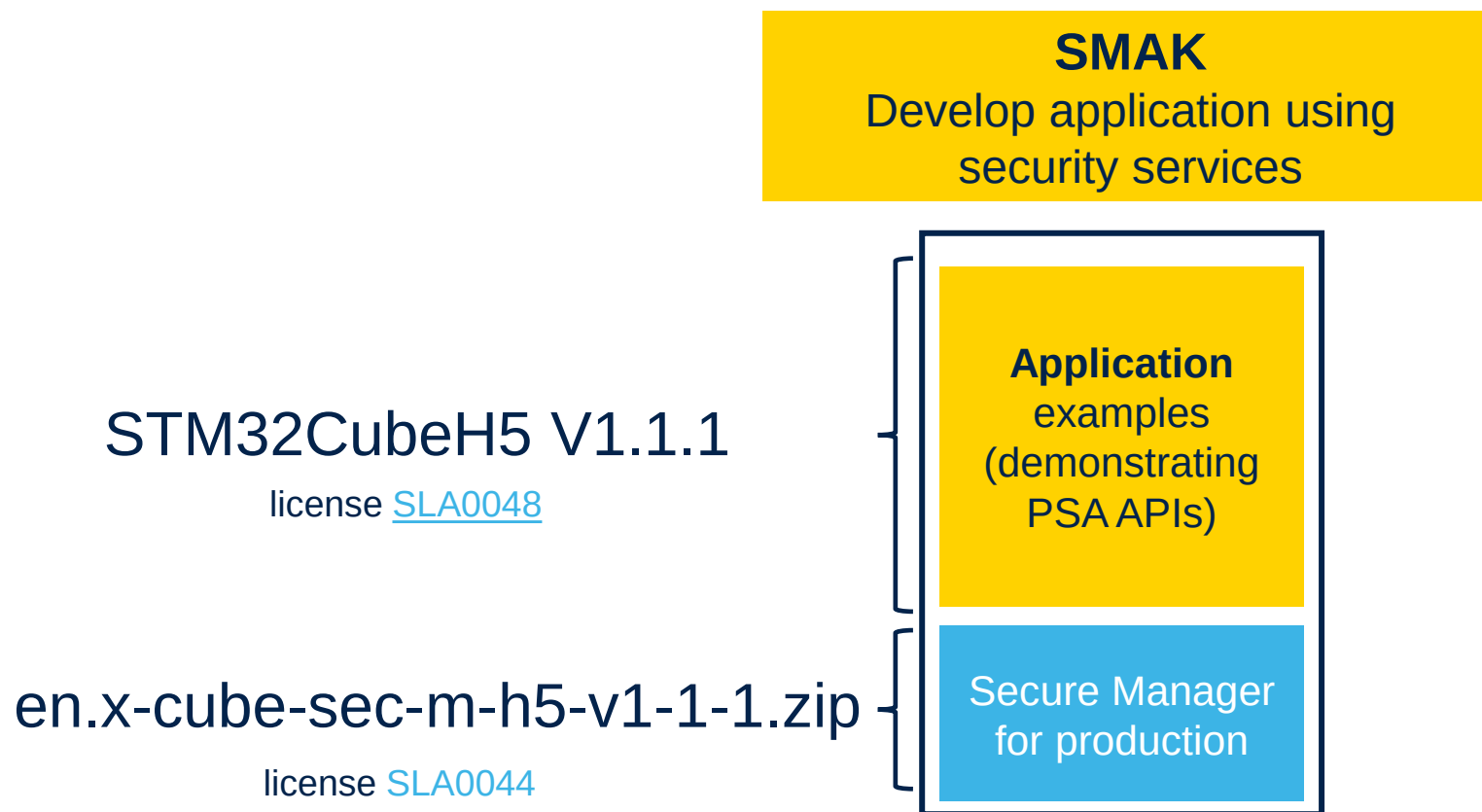
Secure Manager



Secure Manager is provided by ST



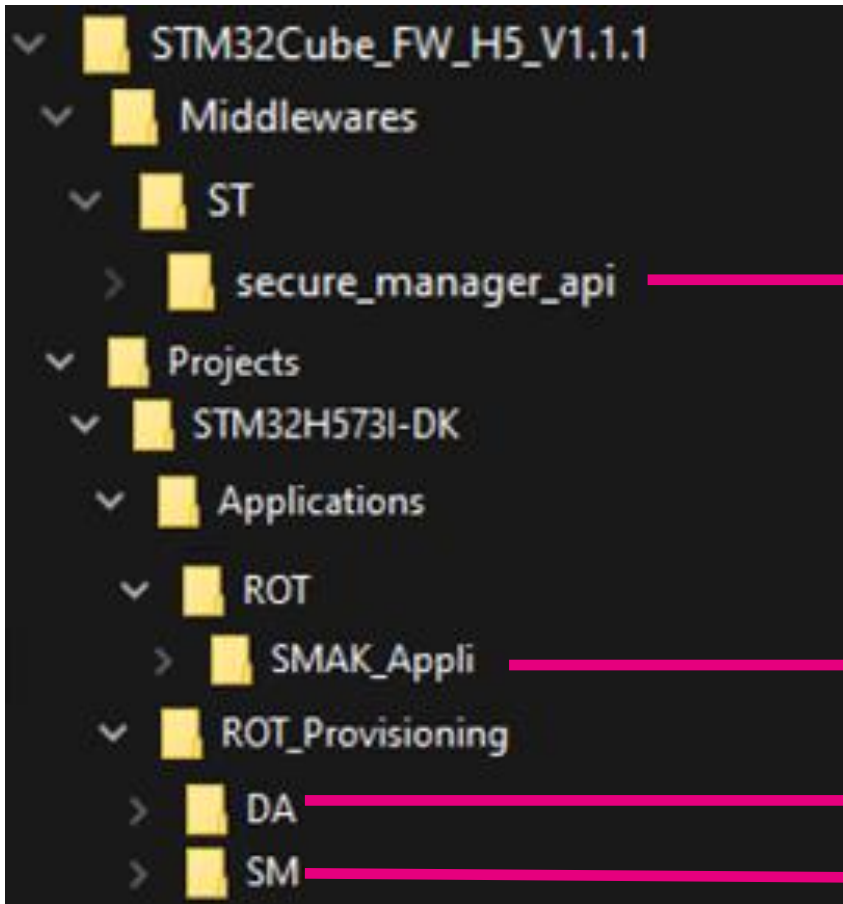
Secure Manager Application Kit (SMAK)





SMAK Delivery overview

Main resources



Secure Manager API Middleware provides callable standard PSA Services

Application example with usage of the secure manager API (PSA API)

Debug Authentication : all the scripts and resource associated

Secure Manager : all the scripts and resource associated



1 Install a Personalized Secure Manager

Initial/virgin state

OEM personalized product

H573

STM32 CubeProgrammer

H573

Default Application

OEM Secrets

Module 2

Module 1

Secure Manager

ST-iRoT

NonSec

Sec

OPEN

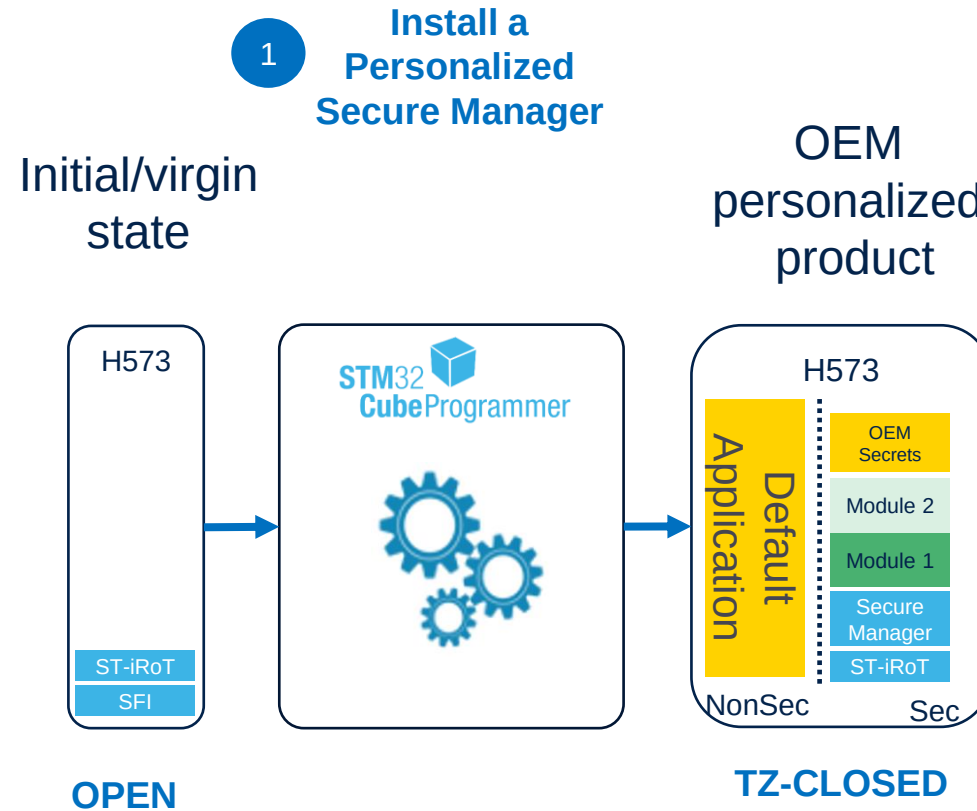
TZ-CLOSED





Secure Manager Ecosystem

Application Development Flow





1 Connect the STM32H5 DK board to the PC

2 STM32Cube_FW_H5_V1.1.1 > Projects > STM32H573I-DK > ROT_Provisioning > SM

Name

- Binary
- Config
- Images
- Keys
- ST

3 provisioning_auto.bat

Once board is connected, run **provisioning_auto.bat**

Wait for end of script => LEDs should blink on the board



Secure Manager installation Use Case

Installation

Step 2 : Installation

```
* BOOT0 pin should be disconnected from VDD
  (STM32H573I-DK: set SW1 to position 0)
  Press any key to continue...
```

```
* Secure installation
  Installation starting (might take up to 15 seconds)...
  Installation completion...
  Successful installation
```

```
====
==== The board is correctly configured with ST Secure Manager Package
====
```

Secure manager installation
securely !

Board LEDs are blinking !



Open UART terminal serial port

The screenshot illustrates the steps to open a UART terminal serial port in Tera Term. The main window is titled "Tera Term - [disconnected] VT". The "File" menu is open, and "New connection..." is highlighted. A dialog box titled "Tera Term: New connection" is shown. The "Serial" radio button is selected. The "Port" dropdown menu is open, showing "COM77: STMicroelectronics STLink Vi" as the selected option. The "OK" button is highlighted.

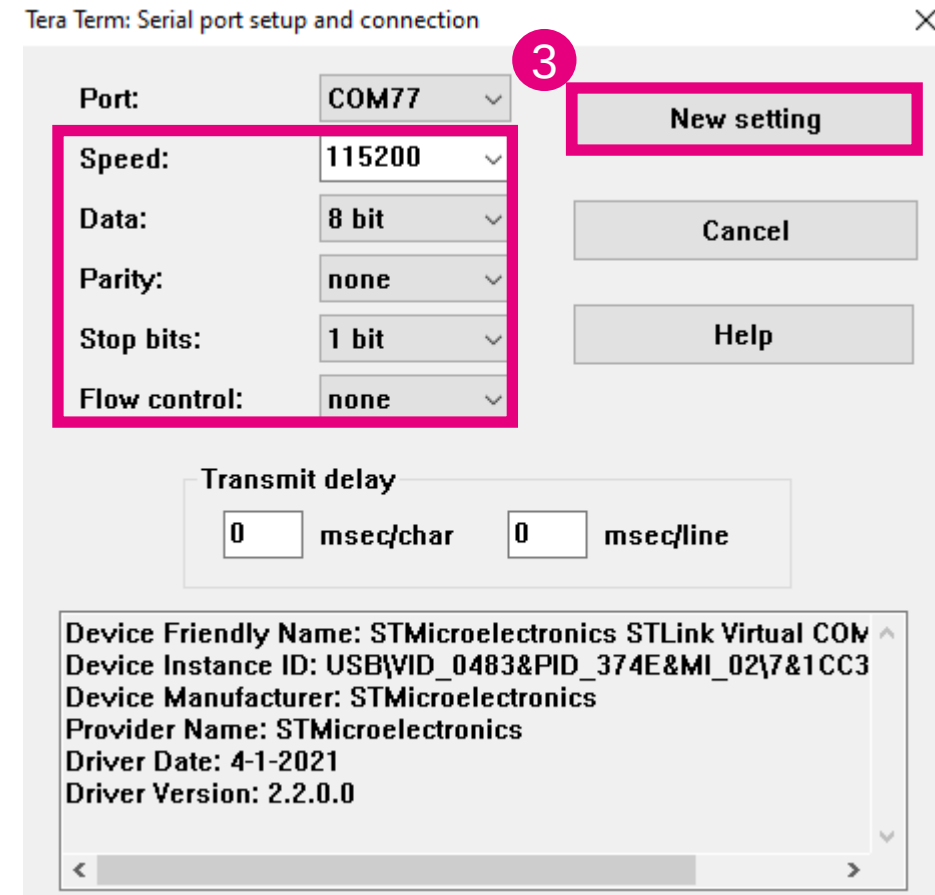
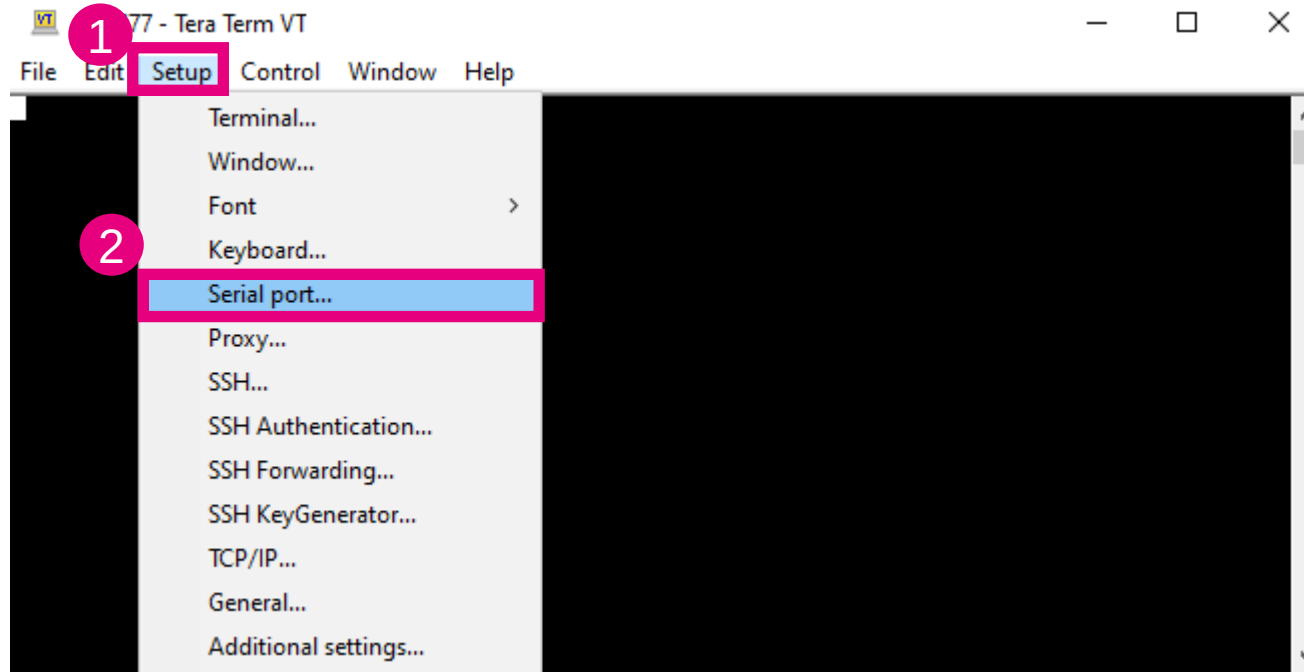
1. File menu, New connection... (Alt+N)

2. Serial connection type, Port: COM77: STMicroelectronics STLink Vi

3. OK button



Serial port configuration





Check terminal connection

- Press reset button on the board

```
COM147 - Tera Term VT
File Edit Setup Control Window Help

==== General Infos =====
Flash Layout configuration index: <0x0>
Secure Manager Version: <1.1.0>
STuRoT Version: <1.1.1>
=====

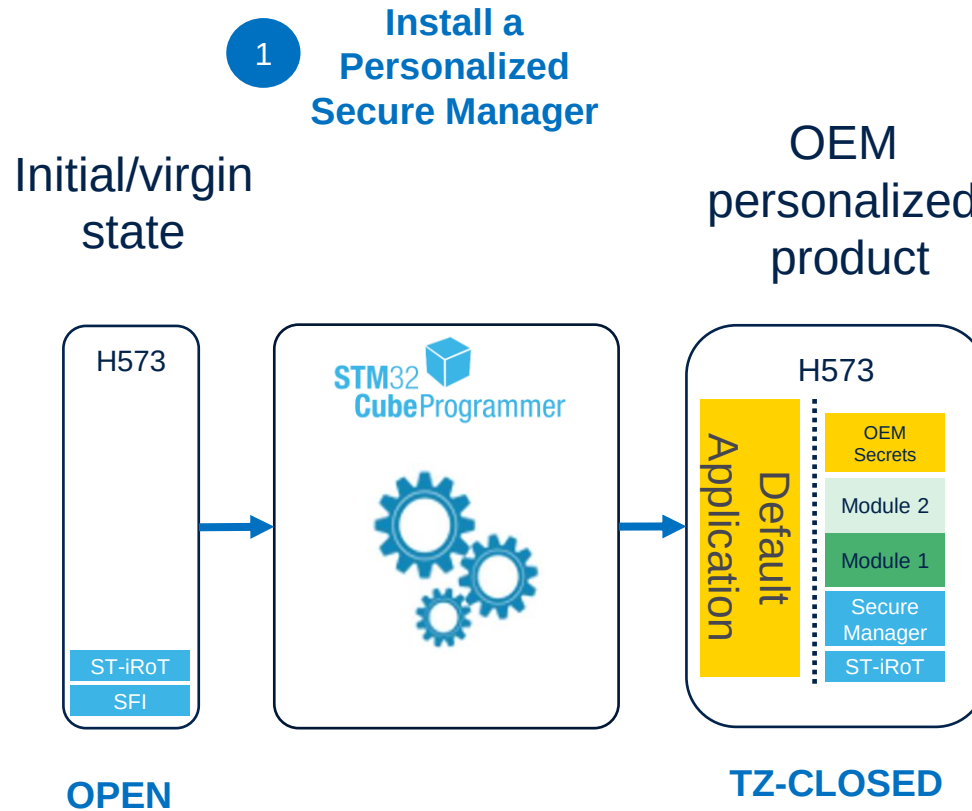
```



Secure Manager Ecosystem

Application Development Flow

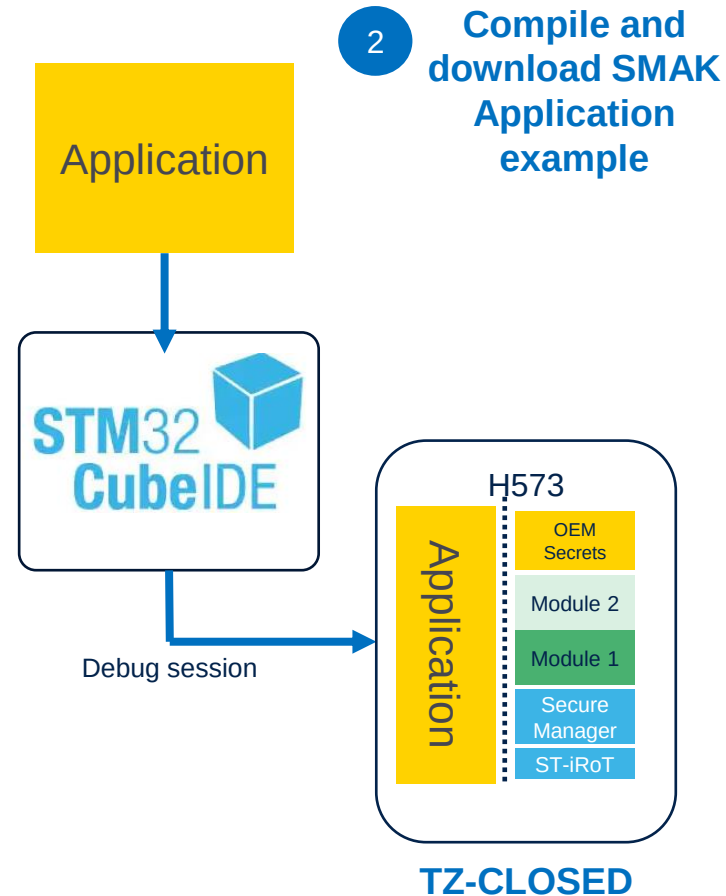
Done





Secure Manager Ecosystem

Application Development Flow





Import SMAK application in the IDE

IDE workspace_1.13.1 - STM32CubeIDE

1 **File** Edit Source Refactor Navigate Search Project Run Window Help myST

2 **Open Projects from File System...**

New Alt+Shift+N >

Open File...

Recent Files >

Close Editor Ctrl+W

Close All Editors Ctrl+Shift+W

Save Ctrl+S

Save As...

Save All Ctrl+Shift+S

Revert

Move...

Rename... F2

Refresh F5

Convert Line Delimiters To >

Print... Ctrl+P

Import...

Export...

Properties Alt+Enter

Switch Workspace >

Restart



Import SMAK application in the IDE

IDE Import Projects from File System or Archive

Import Projects from File System or Archive

This wizard analyzes the content of your folder or archive file to find projects and import them in the IDE.

1

Import source: C:\Users\lefrere\STM32Cube\Repository\STM32Cube_FW_H5_V1.1\Projects\STM32H573I-DK\Applications\ROT\SMAK_Appli\STM32CubeIDE Directory... Archive...

type filter text

Folder	Import as
STM32Cube_FW_H5_V1.1\Projects\STM32H573I-DK\Applications\ROT\SMAK_Appli\STM32CubeIDE	

1 of 1 selected

☐ Hide already open projects

☐ Close newly imported projects upon completion

Use [installed project configurators](#) to:

☒ Search for nested projects

☒ Detect and configure project natures

Working sets

☐ Add project to working sets

Working sets: New... Select...

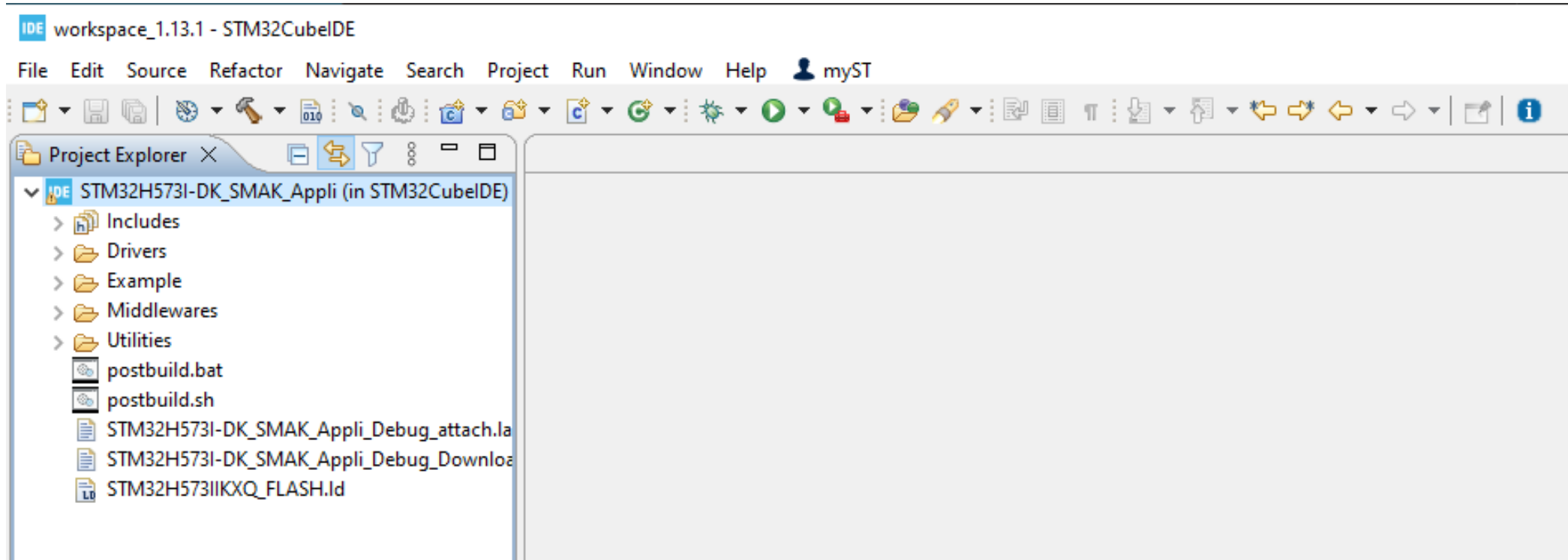
[Show other specialized import wizards](#)

2

< Back Next > **Finish** Cancel

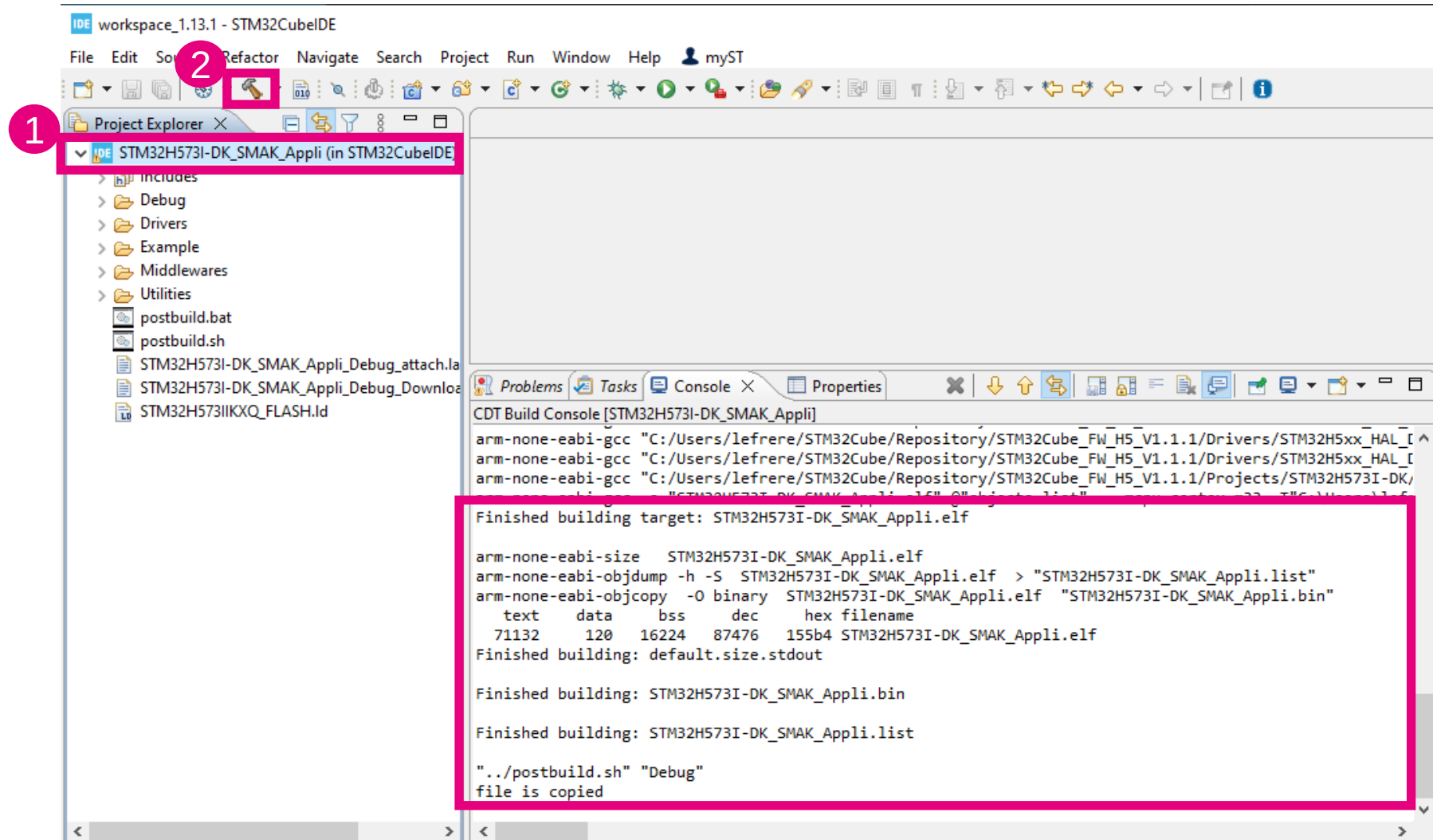


Import SMAK application in the IDE





Compile SMAK application





Load and debug the SMAK application

IDE workspace_1.13.1 - STM32CubeIDE

File Edit Source Refactor Navigate Search Project **Run** Window Help myST

Project Explorer

- IDE STM32H573I-DK_SMAK_Appli (in STM32CubeIDE)
- Includes
- Debug
- Drivers
- Example
- Middleware
- Utilities
- postbuild.bat
- postbuild.sh
- STM32H573I-DK_SMAK_Appli_Debug_attach.la
- STM32H573I-DK_SMAK_Appli_Debug_Download
- STM32H573I-KXQ_FLASH.ld

Run

- Debug
- Run History
- Run As
- Run Configurations...
- Debug History
- Debug As** (selected)
- Debug Configurations...
- Breakpoint Types
 - Toggle Breakpoint
 - Toggle Line Breakpoint
 - Toggle Watchpoint
 - Toggle Method Breakpoint
 - Skip All Breakpoints
 - Remove All Breakpoints
- External Tools

IDE 1 STM32 C/C++ Application

Finished building: STM32H573I-DK_SMAK_Appli.list

"../postbuild.sh" "Debug"

file is copied

postbuild script success

17:28:42 Build Finished. 0 errors, 0 warnings. (took 19s.280ms)

Launch Configuration Selection

Choose a launch configuration to debug

- IDE STM32H573I-DK_SMAK_Appli_Debug_attach
- IDE STM32H573I-DK_SMAK_Appli_Debug_Download_Debug** (selected)

OK Cancel

ST-Link FW update

1 IDE ST-LINK firmware verification

This version of STM32CubeIDE provides a newer firmware version of the attached ST-LINK. Proceed with update?

☒ Always check for latest version.

Yes No

2 STLinkUpgrade 3.12.3

STLINK-V3 Refresh device list

Open in update mode

ST-Link ID: 0046002C3232511739353236

Current Firmware:
Type: STM32 Debug+Mass storage+VCP
Version: Unknown
Update to Firmware: Unknown

3 STLinkUpgrade 3.12.3

STLINK-V3 Refresh device list

Open in update mode

ST-Link ID: 0046002C3232511739353236

Current Firmware:
Type: STM32 Debug+Mass storage+VCP
Version: V3J11M3
Update to Firmware: V3J12M3 STM32 Debug+Mass storage+VCP

Upgrade

4 STLinkUpgrade 3.12.3

STLINK-V3 Refresh device list

Open in update mode

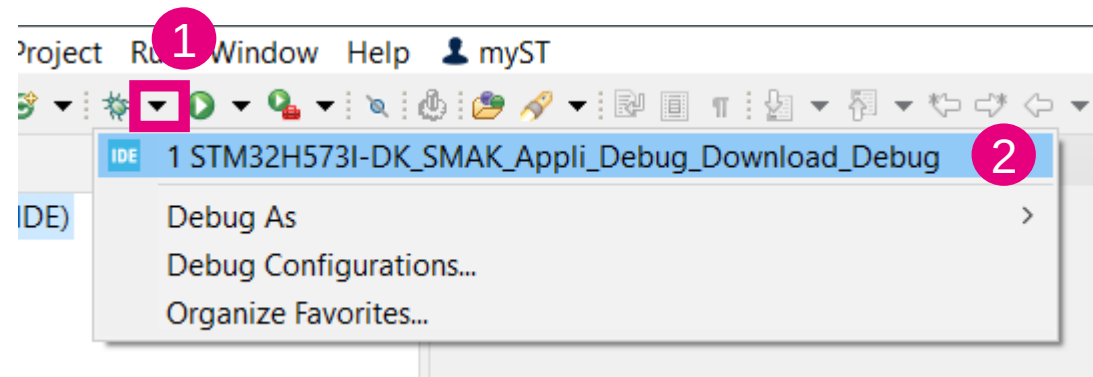
ST-Link ID: 0046002C3232511739353236

Current Firmware:
Type: STM32 Debug+Mass storage+VCP
Version: V3J12M3
Update to Firmware: V3J12M3 STM32 Debug+Mass storage+VCP

Upgrade

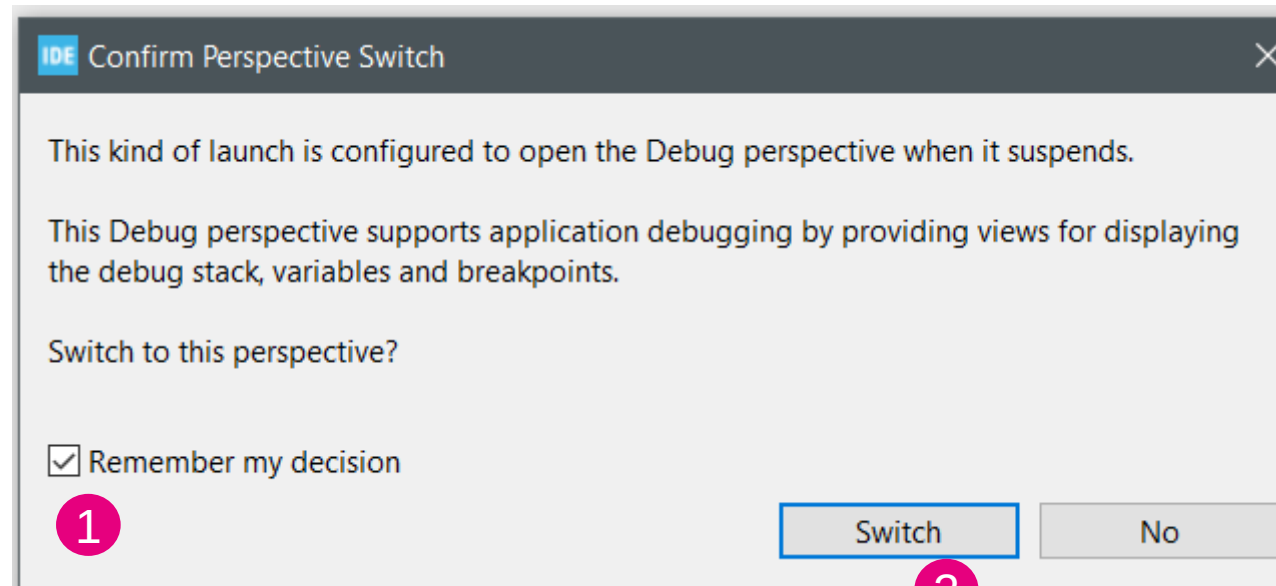
Upgrade successful.

Launch debugger



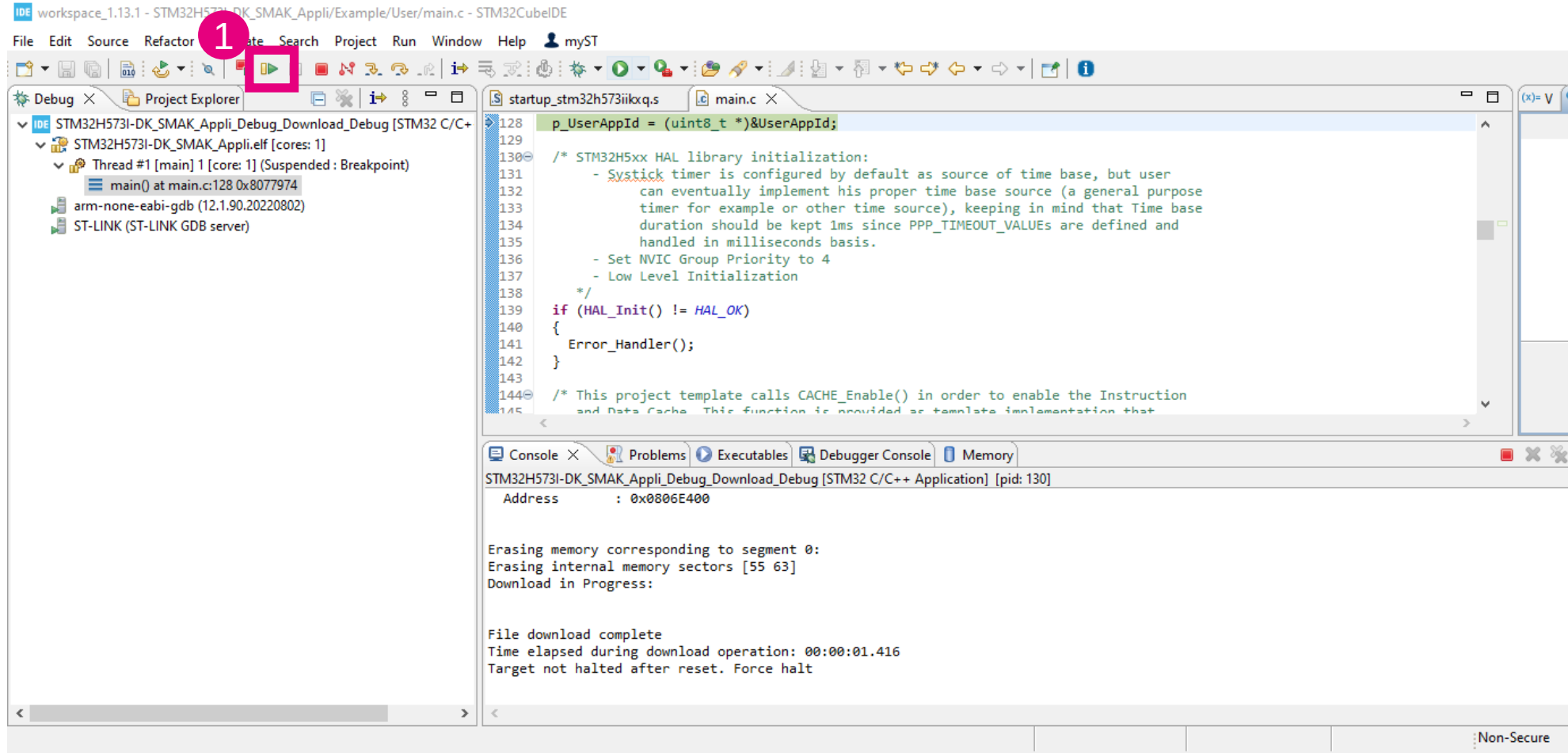


Activate Debug Perspective





Launch the SMAK appli





SMAK appli UI

```
VT COM77 - Tera Term VT
File Edit Setup Control Window Help

=====
<C> COPYRIGHT 2023 STMicroelectronics
=====
Non Secure User App related to Secure Manager #A
=====

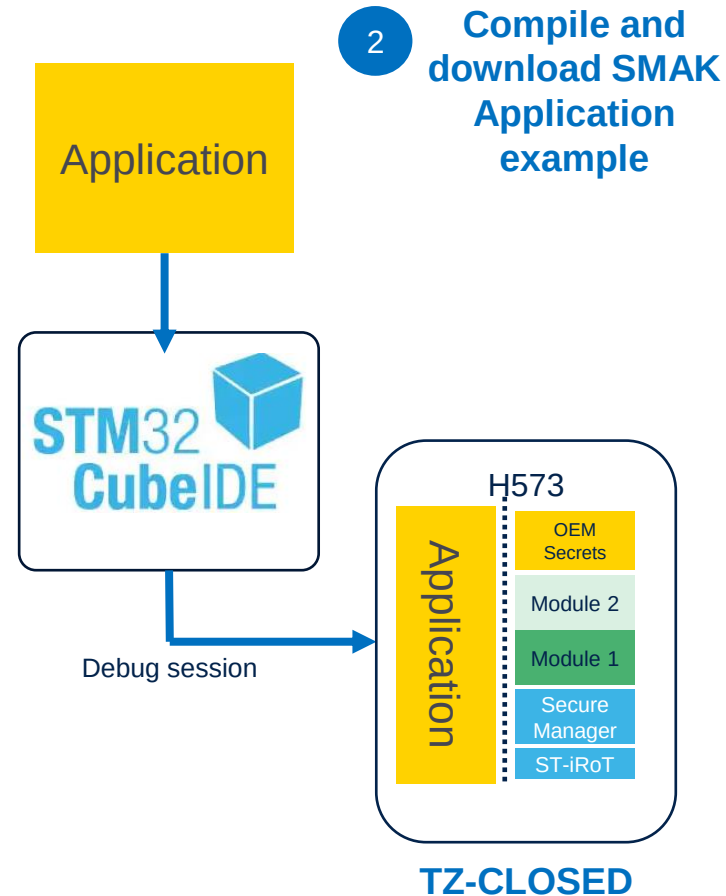
===== Main Menu =====
Internal Trusted Storage ----- 1
Cryptography ----- 2
Initial Attestation ----- 3
Firmware Update ----- 4
Certificates ----- 5
Selection :
█
```



Secure Manager Ecosystem

Application Development Flow

Done





SMAK application UI

```
=====
(C) COPYRIGHT 2023 STMicroelectronics
=====
Non Secure User App related to Secure Manager #A
=====

===== Main Menu =====
Internal Trusted Storage ----- 1
Cryptography ----- 2
Initial Attestation ----- 3
Firmware Update ----- 4
Certificates ----- 5
Selection :
```

- SMAK example application provides implementation of secure services



Internal Trusted Storage

```
===== Internal Trusted Storage Menu =====  
Factory data are provisioned during SM installation and are not updatable  
New data: Set ----- 1  
New data: Get ----- 2  
New data: Get info ----- 3  
New data: Remove ----- 4  
Factory data: Set ----- 5  
Factory data: Get ----- 6  
Factory data: Get info ----- 7  
Factory data: Remove ----- 8  
Previous menu ----- x  
Selection :
```

- Example of dynamic writing into secure store
- Example of accessing pre-provisioned data (Factory data)
- Secure manager provides the ability to pre-provision the Internal Trusted Storage using a simple tool (we will use it later)



Cryptography

```
===== Cryptography Menu =====  
  
RNG ----- 1  
AES GCM ----- 2  
AES CBC ----- 3  
AES CCM ----- 4  
SHA224 ----- 5  
SHA256 ----- 6  
RSA 2048 ----- 7  
ECDSA <DUA USER key> ----- 8  
ECDSA <Factory ITS key> ----- 9  
Previous menu ----- x  
  
Selection :
```

- Set of examples of PSA Crypto API usage



Initial Attestation

```
===== Initial Attestation Menu =====  
Token ----- 1  
Previous menu ----- x  
Selection :
```

- Provides Initial Attestation token used for IOT authentication



Firmware Update

```
===== Firmware Update Menu =====  
Get all applications Status ----- 1  
Install all requested Applications <reboot> ----- 2  
Download Non-Secure App ----- 3  
Request Install Non-Secure App ----- 4  
Accept Non-Secure App ----- 5  
Abort Download Non Secure App ----- 6  
Download Secure Manager ----- 7  
Request Install Secure Manager ----- 8  
Accept Secure Manager ----- 9  
Abort Download Secure Manager ----- A  
Download STuRoT ----- B  
Request Install STuRoT ----- C  
Abort Download STuRoT ----- D  
Previous menu ----- x  
Selection :
```

→ User application secure update

→ Secure manager secure update

→ ST-uROT (2nd stage bootloader) secure update



Certificate

```
===== Certificates Menu =====  
DUA USER X509 certificate ----- 1  
DUA Initial Attestation X509 certificate ----- 2  
Previous menu ----- x  
Selection :
```

- Display both certificate content in X509 pem format

We have seen together :

- how to install the secure manager and how simple it is
- what you can do with secure manager: develop an application and debug it
- an overview of the content of the provided SMAK example application

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